

¹Abdelaziz
Kerbouche

²HamimiChemali

³Mouloud Ayad

Design and efficient implementation of AES system using FPGA



Abstract: - This paper presents efficient implementation of the Advanced Encryption Standard in Field Programmable Gate Array using iterative and pipeline architectures. The research objective is the identification of the optimal design in terms of resource utilization, and timing performance without adjustments to the original algorithm. Pipeline design demonstrates an improved efficiency with high frequency and elevated data processing rate, while iterative implementation was the most resource efficient.

Keywords: Advanced Encryption Standard (AES), Encryption algorithm, Field Programmable Gate Array (FPGA), Embedded systems.

I. INTRODUCTION

The growing demand for connectivity through embedded systems, and maintaining high reliability of interchanged data in employed IoT devices, and above all, the highly valuable processed information, are behind the increase of physical attacks, which have led to the development of appropriate security solutions adapted to such resource-constrained systems [1]. Numerous methods are introduced to resolve data security concern and prevent data hacking, starting from a simple character's substitution technique, and progressing to more complex algorithms that include additional operations [2], [3]. This process of data conversion is called encryption, which is a series of arithmetic operations applied to data before storage or transmission using a secret key that is known exclusively to authorized parties [4]. Two distinct methods for data encryption are commonly used: symmetric and asymmetric. The main distinction between them is the type of key [5]. Symmetric algorithms use the same key for encryption and decryption, while asymmetric algorithms use a pair of public and private keys.

The Advanced Encryption Standard is a symmetric encryption algorithm that replaced DES (Data Encryption Standard) by the U.S NIST (The United States National Institutes of Standard and Technology). AES is widely used for data security due to its efficiency, high level of security and ease of implementation [6]. The major challenge that drew scientist's attention in the AES algorithm is the extended processing time because of the large number of rounds, and the complexity of its integration into resource-limited systems [7]. Accordingly, many research projects are conducted to propose optimization techniques, such as parallel processing, or simplifying some complex tasks and suggesting a lightweight variant of the algorithm.

Two optimization methods for encryption algorithms are commonly used, the first intervenes in the algorithm itself through using a smaller block size, a smaller key size, or go further to removing or altering complex processing modules [8]. The second technique, as in this study, focuses on the optimization of the implementation through reducing critical paths and introducing more parallelism to the design without modifications to the algorithm.

In the last decade, FPGAs are widely deployed in embedded systems due to their flexibility and reconfigurability which offer a convenient environment for embedding computational capabilities to the end user devices [9]. Moreover, hardware implementation enhances area efficiency and limits the processor's role, thus, substantial reduction in power consumption could be achieved.

This project takes advantage of the AES algorithm modularity to synthesize two efficient implementations observing the area occupation and operational timing as monitoring parameters along with the efficiency ratio.

¹*PhD Student, L.C.C.N.S Laboratory, Department of Electronics, University of Ferhat Abbas Sétif 1,
Email:abdelaziz.kerbouche@univ-setif.dz

²Professor, Department of Electronics, University of Ferhat Abbas Sétif 1, Email:hamimi.chemali@univ-setif.dz

³ Professor, Department of Electronics, University of Ferhat Abbas Sétif 1, Email:mouloud.ayad@univ-setif.dz

Proposed designs are implemented on Vertex-7 FPGA, and results reveal notable distinctions between developed approaches, where the pipeline system succeeded to attain a high frequency of 434.78 MHz with a corresponding throughput of about 55.65 Gbps and efficiency ratio of 5.87 compared to 364.17 MHz, 4.66 Gbps and 3.51 efficiency ratio in iterative design.

II. APPROACH STUDY

AES is a thoroughly described algorithm in the literature [10]. It is a subset of the Rijndael algorithm with a data block size of 128-bits and a cipher key length of 128, 192 or 256 bits [6]. AES uses a series of mathematical operations in two separated routines, the encryption process and the Key generation.

The Encryption process consists of repeated iterations known by Rounds [11], each round is composed of sequential operations called SubBytes, ShiftRows, MixColumns and AddRoundKey. The Key generation is the process of expansion of the initial key to generate partial keys for each round. The overall AES algorithm is presented in Fig. 1.

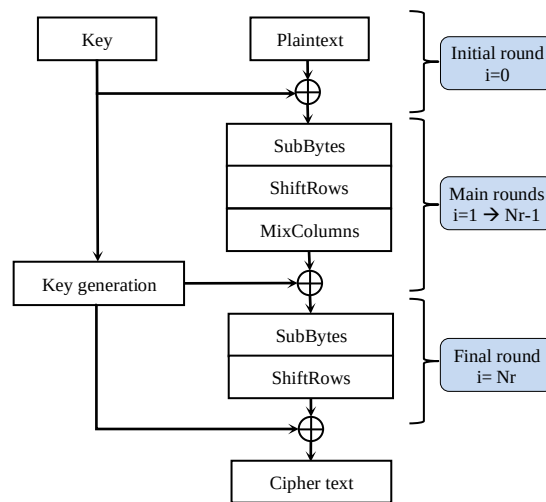


Fig. 1 AES algorithm flowchart.

Iterative implementation is a closed loop structure of the main operations interconnected as point-to-point blocks in a feedback loop, the same blocks are therefore used during all the rounds [15]. This technique requires an additional unit to manage dataflow as shown in Fig. 2.

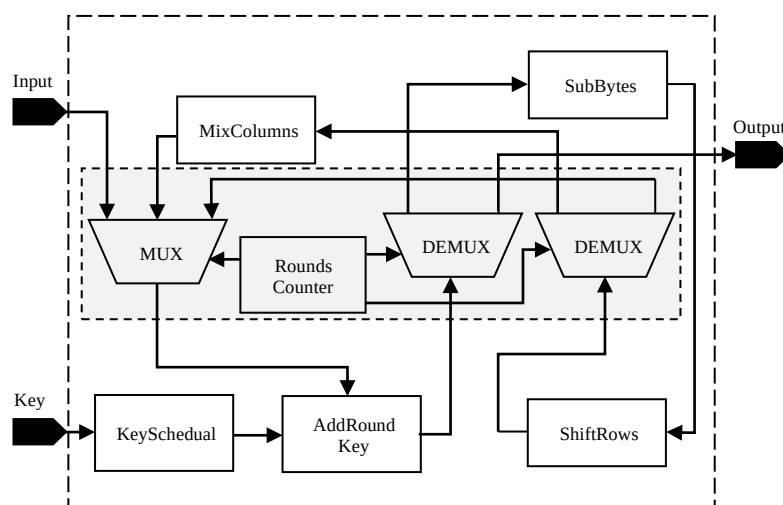


Fig. 2 Iterative implementation structure.

The major limitation of this method is the idle time caused by the iterations effect, but it requires less resources compared to other techniques, which makes it more compatible with resource-limited systems [16]. The iterative system process is illustrated in Fig. 3.

Pipeline structure consists of multiple AES modules interconnected sequentially [17]. Each module processes one round, hence, it operates one time during the encryption process and becomes free for the next input. As pipeline architecture allows processing several inputs for reducing idle time and critical path, it increases the operating frequency, but consumes more resources. Fig. 4 illustrates the AES pipeline architecture, and Fig. 5 shows the pipeline process. This technique is well adapted to systems that require a steady stream of data.

	Add round-key	Sub bytes	Shift rows	Mix columns
Round 0	Processing	IDLE	IDLE	IDLE
Round 1	IDLE	Processing	IDLE	IDLE
	IDLE	IDLE	Processing	IDLE
	IDLE	IDLE	IDLE	Processing
	Processing	IDLE	IDLE	IDLE
Round 2	IDLE	Processing	IDLE	IDLE
	IDLE	IDLE	Processing	IDLE
	IDLE	IDLE	IDLE	Processing
	Processing	IDLE	IDLE	IDLE
⋮				
Round 10	IDLE	Processing	IDLE	IDLE
	IDLE	IDLE	Processing	IDLE
	Processing	IDLE	IDLE	IDLE

Fig. 3 The functional process of the iterative system

SubBytes is a nonlinear operation, in which each byte in the input is transformed as a polynomial in the Galois Field $GF(2^8)$ using the affine transformation and matrix multiplication [12]. The same results can be obtained using a precalculated substitution table called S-box.

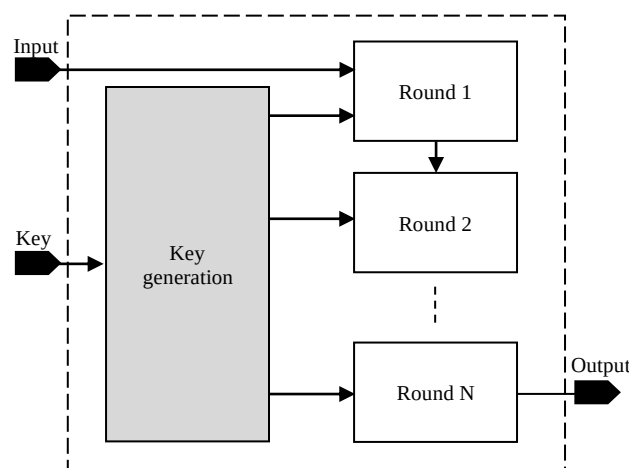


Fig. 4 The pipeline architecture of AES algorithm

ShiftRows and MixColumns are linear transformations to the plaintext, where rows are shifted and columns are mixed in an invertible manner. Rows shifting operation consists of sliding rows to the left by a given offset, the first row is unchanged, while the second row is shifted over one byte, similarly the third and the last rows are shifted by two and three bytes respectively. In the same way, MixColumns changes the order of columns by multiplying the plaintext by a given matrix.

AddRoundKey is the operation where the main process meets the key generation process. It is considered the most important step since it mixes the plaintext with the secret key to hides the relationship between the original data and the ciphertext [13].

The set of operations described earlier are repeated N times in a feedback loop, where N is the number of rounds. The input to each round during the encryption process is driven from the output of the previous round in a closed loop structure, which is a point of strength in terms of security [14], but also a big challenge for resources-limited and interactive systems. Numerous methodologies of AES implementation have been elaborated, and this work focuses on iterative and pipeline techniques.

Proposed architectures are designed using a Hardware Description Language (HDL). Synthetization and simulation results are compared to similar works in terms of resource utilization, throughput, and efficiency ratio. Fig. 6 shows the development process of this project.

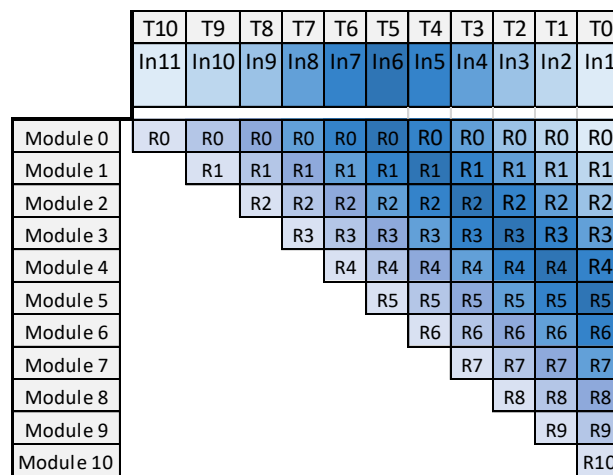


Fig. 5 The functional process of the pipeline system

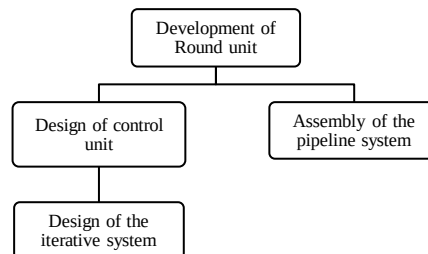


Fig. 6 The development process of proposed architectures

A. The development of round unit

The AES-round unit includes main operations: SubBytes, ShiftRows, MixColumns and AddRoundkey, in addition to the KeyGeneration process. The AES-round unit is the fundamental element of the overall system. The schematic of the round module is shown in Fig. 7.

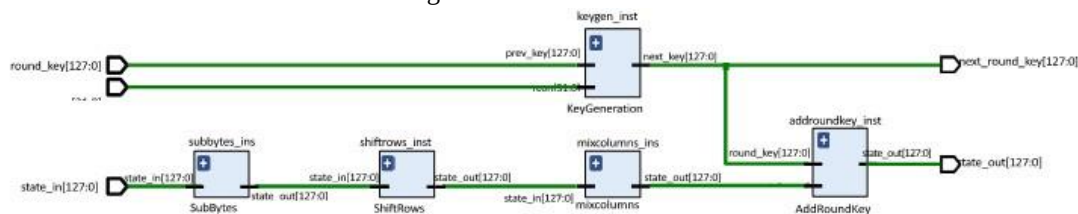


Fig. 7 The schematic of the round module

B. Design of the control unit

The rounds controlling unit plays a key role in timing performance and reducing latency through managing rounds counting, dataflow, and synchronize between modules.

C. Design of the iterative system

The assembly of the round unit with the controlling unit in a closed loop structure forms the iterative system. The schematic of the iterative system is presented in Fig. 8.

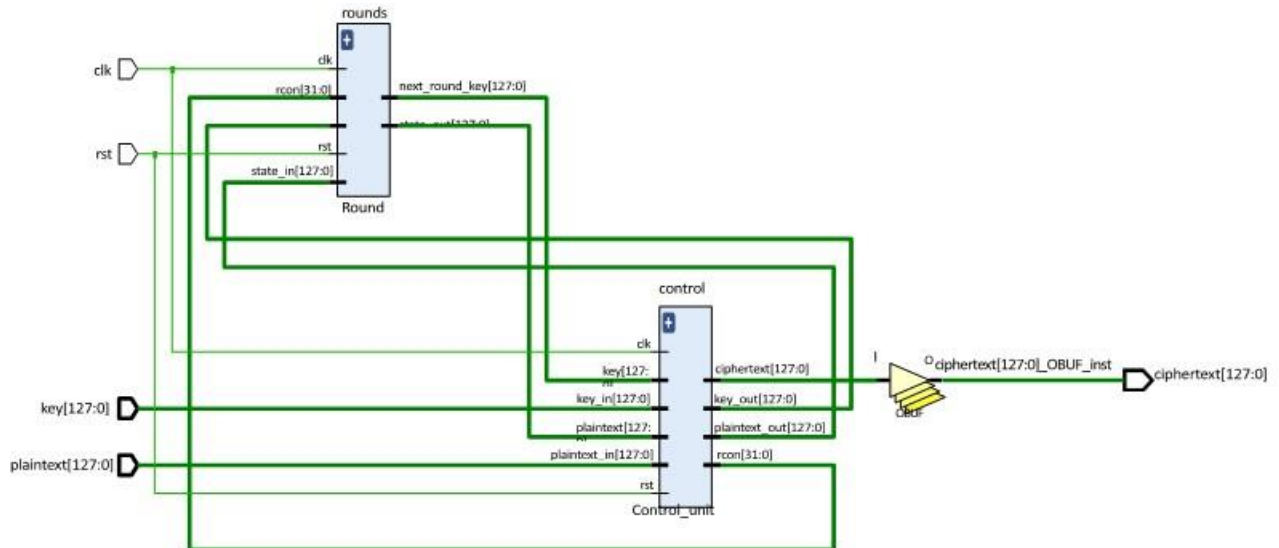


Fig. 8 The schematic of the iterative system.

D. The Implementation of the pipeline system

Pipeline implementation is designed through placing 10 round units in serial, each round unit operates one single time during the same encryption process, as a result, the system processes a new input each clock cycle. the schematic diagram of pipeline implementation is illustrated in Fig. 9.

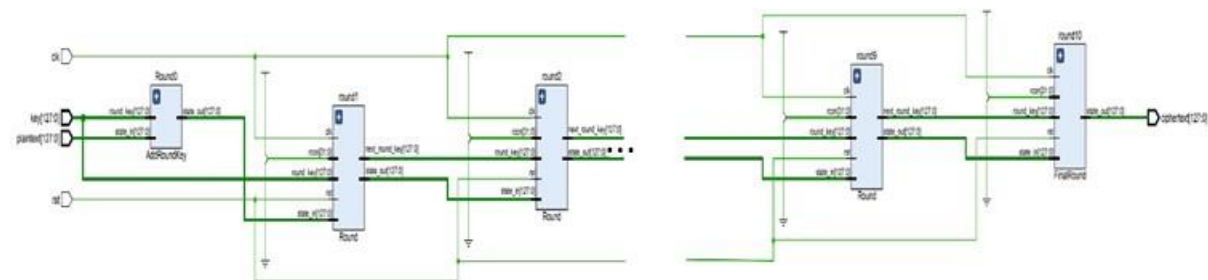


Fig. 9 The schematic of the pipeline system.

III. VALIDATION

Graph in Fig. 10 illustrates resource utilization, throughput and efficiency ratio of the proposed systems.

We notice that pipeline architecture processes data faster at the expense of using more resources than the iterative system. The progression of efficiency ratio when transitioning from iterative to pipeline systems provides a distinct indication of the optimal trade-off between resource utilization and operational time.

Table 1 summarizes timing results of the methods proposed in this work, and a comparison with similar works elaborated in [18], [19], [20], [21] and [22] in terms of resource utilization and efficiency ratio.

Obtained results highlight the importance of choosing a convenient configuration according to the resources availability and timing criticality of the designed system.

The proposed pipeline architecture as shown in Table 1 achieved the highest operating frequency and outperformed in terms of data processing rate compared to similar pipeline and sub-pipeline works presented in [20], [19], and [18] due to the minimal critical path delay. Iterative design reaches the topmost efficiency as it requires less resources compared to similar non-pipelined works although it operates at a lower frequency and a slightly inferior throughput against the project presented in [22].

Tab. 1. Comparison of the proposed architectures results with published works.

Architecture	Device	Number of clock cycles	Frequency (MHz)	Slices	Throughput (Gbps)	Efficiency (Mbps/Slice)
Pipeline system [18]	Spartan-3	12	-	46745	0.75 (Cycles/Byte)	-
Pipeline system [19]	Virtex-6	-	190.65	1551	0.56	0.361
AES- MPPRM [20]	Virtex-6	1	433.28	8342	-	-
Pipeline system (proposed)	Virtex-7	1	434.78	9466	55.65	5.87
AES-CTR [21]	Virtex-6	11	318.67	7758 (area)	20.3	2.85
Non-Pipelined [22]	Virtex-7	11	456	2444	5.30	-
Iterative system (proposed)	Virtex-7	10	364.17	1329	4.66	3.51

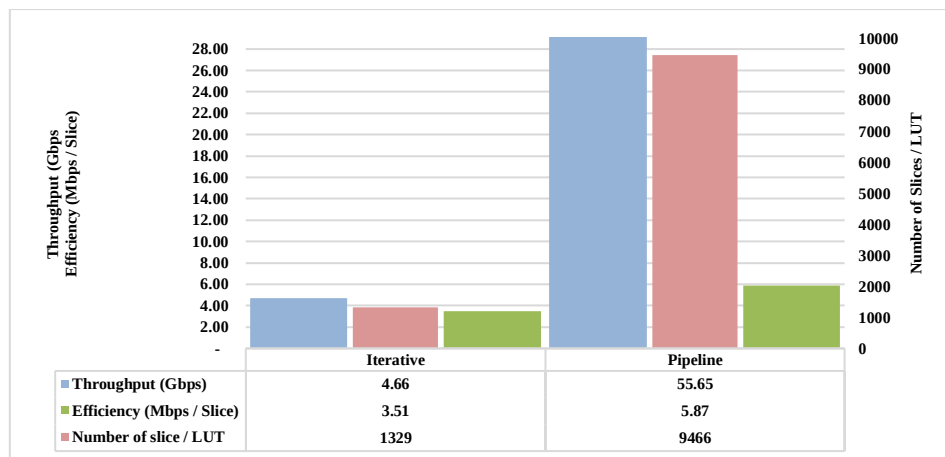


Fig. 10. Graphical representation of resource utilization, throughput, and efficiency ratio for the proposed systems.

IV. SIMULATION RESULTS

Simulation results of iterative and pipeline implementations are shown in Fig. 11 and Fig. 12 respectively.

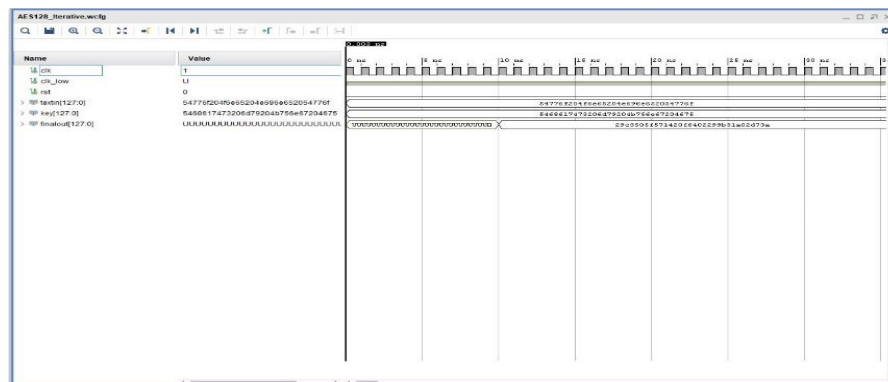


Fig. 11 Simulation results of the iterative system

The encryption duration of the iterative system is 10 clock cycles with maximum frequency of 364,17 MHz and a throughput of 4,66 Gbps.

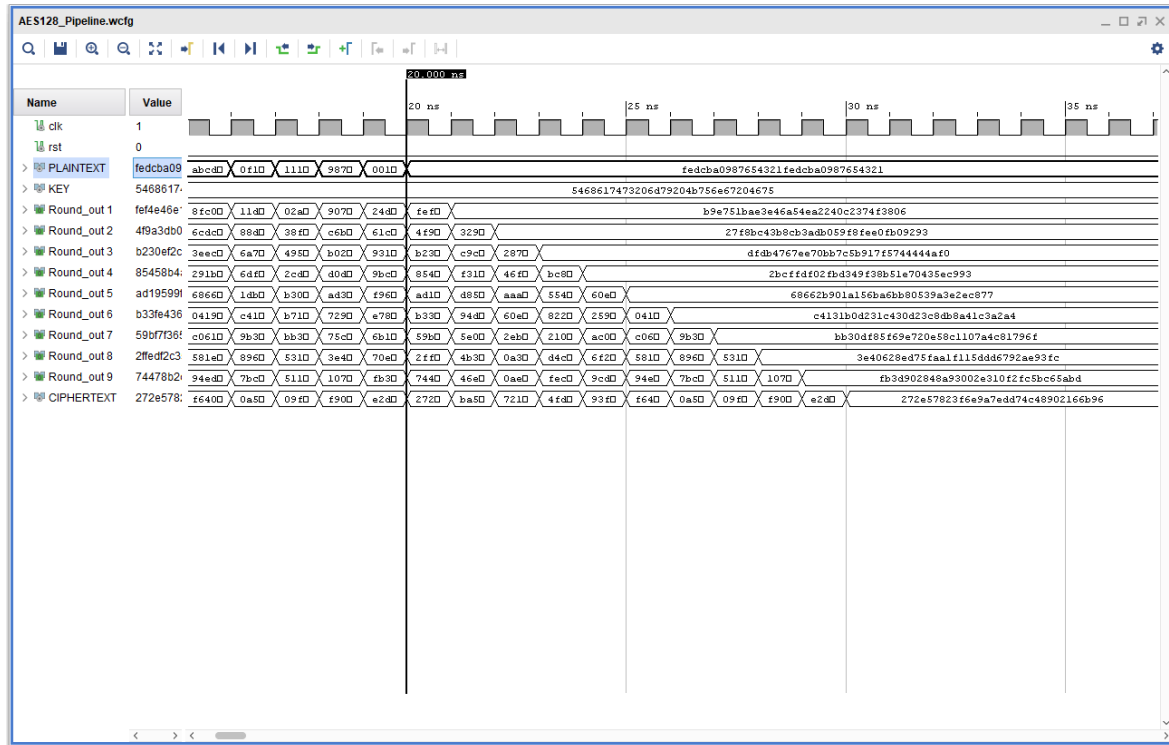


Fig. 12 Simulation results of the pipeline system

Pipeline architecture processes the first input in 10 clock cycles, and generates following outputs each clock cycle with a frequency of 434,78 MHz and a throughput of 55,65 Gbps as shown in Fig. 12. The high frequency is achieved due to reduced latency resulted from parallel processing.

V. CONCLUSION

The impact of security implementation method as far as resource utilization and timing performance are concerned, is studied and may provide a quantification criterion for embedded systems. In a future work, the introduction of partial reconfiguration will be considered to improve embedded systems robustness and efficiency while maintaining a high level of security.

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